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GENERATING AN AUDIO SIGNAL FROM MULTIPLE INPUTS

Abstract

A system, such as an ear-wearable device or a hearing aid, can receive multiple audio signals representing a same audio content, can cross-correlate the multiple audio signals to determine relative delays between the audio signals, can apply the determined delays to at least one of the audio signals to form multiple synchronized audio signals, and can mix at least two of the synchronized audio signals in time-varying proportions to form an output audio signal. The system can optionally adjust the mix proportions, in real time, to increase or optimize the signal-to-noise ratio of the output audio signal. The system can optionally perform the cross-correlation repeatedly, at regular or irregular time intervals, to update the relative delays. The system can optionally divide the audio signals into frequency bands, and apply these operations to each frequency band, independent of the other frequency bands.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 17/755,248, filed Apr. 25, 2022, which is a U.S. National Stage Filing under 35 U.S.C. 371 from International Application No. PCT/US2020/070729, filed on Oct. 30, 2020, and published as WO 2021/087524, which claims the benefit of U.S. Provisional Application No. 62/927,961, filed Oct. 30, 2019, each of which are hereby incorporated by reference in their entirety.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates generally to improving signal quality in an audio system that can use multiple sources for an audio signal.

BACKGROUND OF THE DISCLOSURE

[0003] There have been many advances in ear-wearable devices, including hearing aids. Early hearing aids included a microphone to convert sound waves (e.g. time-varying pressure that represents an audio signal) to an electrical signal, an amplifier to process the electrical signal, and a receiver to present the processed signal to a listener's ear. Recent advances in hearing aids can additionally allow a listener to switch from the microphone to alternate sources for the audio signal.

[0004] For example, some modern hearing aids can allow the listener to switch between the microphone and a signal received by a telecoil. A telecoil can include a loop of electrically conductive wire positioned in the hearing aid and configured to receive signals from external wire loops via a magnetic field (e.g., via Faraday's law of induction). So-called loop systems can provide audio content in these external wire loops. A loop system can include a wire loop that encircles a particular area, such as a seating area of a theater, or a back seat of a taxicab.

[0005] As another example, some modern hearing aids can include an antenna that can receive audio content through a specified wireless digital protocol. For example, a modern hearing aid can couple to a wireless stream of an audio source via Bluetooth connectivity.

[0006] Some modern hearing aids can switch automatically between audio sources, based on events. For example, if a magnetic switch in the hearing aid detects a magnetic field produced by a telephone handset, then the hearing aid can automatically switch to the telecoil. As another example, if a radio in the hearing aid detects a presence of a wireless stream via an antenna in the hearing aid, the hearing aid can automatically switch to the wireless stream.

[0007] One drawback to such automatic switching is that it may not improve a quality of the processed signal sent to the listener's ear. For example, in a case where a microphone signal may be relatively clean, but a corresponding Bluetooth signal may be relatively noisy, such automatic switching may result in the relatively noisy signal being processed and directed to the listener's ear.

[0008] There is ongoing effort to improve the quality of the processed signal sent to the listener's ear.

SUMMARY

[0009] In an example, a system for generating an audio signal from multiple inputs can include: at least one processor and memory coupled to the at least one processor. The memory can store instructions that, when executed by the at least one processor, cause the at least one processor to execute operations. The operations can include: receiving a first audio signal and a second audio

signal that both represent a same audio content; cross-correlating the first and second audio signals to determine a relative delay between the first and second audio signals; applying the determined delay to at least one of the first or second audio signals to form a first synchronized audio signal representing the first audio signal and form a second synchronized audio signal representing the second audio signal, the first synchronized audio signal being synchronized with the second synchronized audio signal; and mixing the first and second synchronized audio signals in time-varying proportions to form an output audio signal that represents the audio content.

[0010] In an example, a method for generating an audio signal from multiple inputs can include: receiving a first audio signal and a second audio signal that both represent a same audio content; cross-correlating the first and second audio signals to determine a relative delay between the first and second audio signals; applying the determined delay to at least one of the first or second audio signals to form a first synchronized audio signal representing the first audio signal and form a second synchronized audio signal representing the second audio signal, the first synchronized audio signal being synchronized with the second synchronized audio signal; and mixing the first and second synchronized audio signals in time-varying proportions to form an output audio signal that represents the audio content.

[0011] In an example, an ear-wearable device can include: a housing; at least one processor disposed in the housing; a microphone coupled to the at least one processor and configured to convert sound waves proximate the housing to a microphone audio signal that represents an audio content; a telecoil coupled to the at least one processor and configured to convert a modulated electromagnetic field proximate the housing to a telecoil audio signal that represents the audio content; an antenna coupled to the at least one processor and configured to convert a wireless signal proximate the housing to a wireless audio signal that represents the audio content; and memory coupled to the at least one processor. The memory can store instructions that, when executed by the at least one processor, cause the at least one processor to execute operations. The operations can include: receiving the microphone audio signal, the telecoil audio signal, and the wireless audio signal; cross-correlating the microphone audio signal, the telecoil audio signal, and the wireless audio signal to determine relative delays among the microphone audio signal, the telecoil audio signal, and the wireless audio signal; applying the determined delays to at least some of the microphone audio signal, the telecoil audio signal, or the wireless audio signal to form a synchronized microphone audio signal representing the microphone audio signal, form a synchronized telecoil audio signal representing the telecoil audio signal, and form a synchronized wireless audio signal representing the wireless audio signal, the synchronized microphone audio signal, the synchronized telecoil audio signal, and the synchronized wireless audio signal being synchronized to one another; and mixing at least two of the synchronized microphone audio signal, the synchronized telecoil audio signal, or the synchronized wireless audio signal in time-varying proportions to form an output audio signal that represents the audio content. The hearing aid can include a speaker disposed in the housing and configured to produce audio corresponding to the output audio signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] FIG. 1 shows an example of a system for generating an audio signal from multiple inputs, in accordance with some examples.

[0013] FIG. 2 shows an example of cross-correlation data from the system of FIG. 1, in accordance with some examples.

[0014] FIG. 3 shows a schematic diagram of an example of a system for generating an audio signal from multiple inputs, in accordance with some examples.

[0015] FIG. 4 shows a flow chart of a method for generating an audio signal from multiple inputs, in accordance with some examples.

[0016] Corresponding reference characters indicate corresponding parts throughout the several views. Elements in the drawings are not necessarily drawn to scale. The configurations shown in the drawings are merely examples and should not be construed as limiting in any manner.

DETAILED DESCRIPTION

[0017] The systems and methods discussed herein have multiple applications, including use in ear-wearable devices (e.g., a hearing aid, headphones, etc.). Although the use in ear-wearable devices such as a hearing aid is discussed in detail herein, it will be understood that the systems and methods discussed herein can also apply beyond ear-wearable devices to other devices such as sound mixing boards at concert venues, and other suitable applications.

[0018] When used in a hearing aid of a listener, the systems and methods discussed herein can apply to many different situations. As a first example, the systems and methods discussed herein can apply to a listener positioned in a room with a loop system, with a telecoil in the hearing aid, and with an acoustic signal of interest in the room. As a second example, the systems and methods discussed herein can apply to a listener streaming audio to the hearing aid from a television, telephone, remote microphone, or other device, additionally with an acoustic signal of interest in the room. As a third example, the systems and methods discussed herein can apply to a listener having an acoustic signal of interest in the room, and additionally having a companion microphone that also picks up the acoustic signal in the room. As a fourth example, the systems and methods discussed herein can apply to a listener having an accelerometer and/or a vibration sensor in the hearing aid, where the listener wants to attenuate the sound of his or her own voice (thereby reducing an occlusion effect) or enhance his or her own voice (such as, on a phone call). As a fifth example, the systems and methods discussed herein can apply to a listener participating in a phone call in microphone mode or telecoil mode without realizing there is a good signal on another transducer. In all of these five examples, the systems and method discussed herein can increase or maximize a signal-to-noise ratio of the output audio signal, which can improve the sound for the listener.

[0019] FIG. 1 shows an example of a system **100** for generating an audio signal from multiple inputs, in accordance with some examples. In the example of FIG. 1, the system **100** is configured a hearing aid. It will be understood that other configurations are also possible, such as for an audio mixer or mixing board that can combine signals from multiple microphones.

[0020] The system **100** can receive multiple audio signals representing a same audio content. In this example, the audio content is the audio from a television program. In this example, the audio signals are an acoustic signal **102** originating from a loudspeaker **104**, a wireless signal **106** originating from an audio streamer **108**, and a magnetic signal **110** originating from a room loop system **112**. In this example, the acoustic signal **102** is delayed by 10 milliseconds, due to the acoustic sound waves propagating 3.4 meters (11 feet) in air at the speed of sound (343 meters per second) from the loudspeaker **104** to the listener. In this example, the wireless signal **106** is delayed by a relatively large 40 milliseconds, due to the time required to process data packets at the audio streamer **108**. In this example, the magnetic signal **110** is delayed by a relatively small 3 milliseconds, due to the relatively fast analog processing in the room loop system **112**.

[0021] The system **100** can cross-correlate the multiple audio signals to determine relative delays between the audio signals, can apply the determined delays to at least one of the audio signals to form multiple synchronized audio signals, and can mix at least two of the synchronized audio signals in time-varying proportions to form an output audio signal. Synchronizing the audio signals in this manner can allow the system **100** to switch among the audio signals and/or combine the audio signals as needed. This can help avoid artifacts that might arise from switching between audio signals that are out of synchronization, such as glitches or small pauses. This can also help avoid artifacts that might arise from combining audio signals that are out of synchronization, such

as having a tinny tonal quality, smearing of audio, or a ringing or delay.

[0022] The system **100** can optionally adjust the mix proportions, in real time, to increase or maximize a signal-to-noise ratio of the output audio signal, which can raise or optimize the signal-to-noise ratio of the output audio signal. The system **100** can optionally perform the cross-correlation repeatedly, at regular or irregular time intervals, to update the relative delays. The system **100** can optionally divide the audio signals into frequency bands, and apply these operations to each frequency band, independent of the other frequency bands.

[0023] FIG. **2** shows an example of cross-correlation data from the system **100** of FIG. **1**, in accordance with some examples. In FIG. **2**, the horizontal axis corresponds to time or delay, and the vertical axis corresponds to (dimensionless) correlation value.

[0024] Element **202** is a specified positive correlation threshold value. Element **204** is a specified negative correlation threshold value. If a positive peak in a cross-correlation data set (curve) exceeds the positive correlation threshold value **202**, or a negative peak is lower than the negative correlation threshold value **204**, the system **100** deems the two correlated audio signals to be suitably correlated, and therefore deems that the two correlated audio signals correspond to the same audio content.

[0025] FIG. **2** shows three cross-correlation data sets, or curves, which correspond to the number of ways the three audio signals of FIG. **1** can be combined pairwise.

[0026] A first curve **206** (dotted) shows the cross-correlation between the acoustic signal **102** (“mic”) and the magnetic signal **110** (“coil”). The first curve **206** has a positive peak value **208** that exceeds the positive correlation threshold value **202**. If there are both negative and positive peaks, the peak having the highest absolute value is used. The location **210** of the peak value **208** represents the delay between the acoustic signal **102** (“mic”) and the magnetic signal **110** (“coil”). In this example, the acoustic signal **102** lags the magnetic signal **110** by 7 milliseconds.

[0027] A second curve **212** (dashed) shows the cross-correlation between the acoustic signal **102** (“mic”) and the wireless signal **106** (“stream”). The second curve **212** has a positive peak value **214** that exceeds the positive correlation threshold value **202**. The location **216** of the peak value **214** represents the delay between the acoustic signal **102** (“mic”) and the wireless signal **106** (“stream”). In this example, the acoustic signal **102** leads the wireless signal **106** by 30 milliseconds.

[0028] A third curve **218** (solid) shows the cross-correlation between the magnetic signal **110** (“coil”) and the wireless signal **106** (“stream”). The second curve **218** has a negative peak value **220** that is less than the negative correlation threshold value **204**. Because the peak value **220** is negative, the magnetic signal **110** or the wireless signal **106** and the system **100** are out of phase with each other. To correct the phase, the system **100** can reverse the polarity of the magnetic signal **110** or the wireless signal **106**, so that when the magnetic signal **110** or the wireless signal **106** are summed, they add, rather than cancel. The location **222** of the peak value **220** represents the delay between the magnetic signal **110** (“coil”) and the wireless signal **106** (“stream”). In this example, the magnetic signal **110** leads the wireless signal **106** by 37 milliseconds.

[0029] In this example, the delay values of 7 milliseconds, 30 milliseconds, and 37 milliseconds can be below a specified delay threshold value, such as 50 milliseconds. In some examples, the system **100** can check to ensure that the delay values are less than the specified delay threshold value to continue the processing.

[0030] FIG. **3** shows a schematic diagram of an example of a system **300** for generating an audio signal from multiple inputs, in accordance with some examples. The system **300**, as shown in the example of FIG. **3** is configured as a hearing aid. It will be understood that other configurations are also possible, such as for an audio mixer or mixing board that can combine signals from multiple microphones. In some examples, the system **300** can receive multiple audio signals representing a same audio content, can cross-correlate the multiple audio signals to determine relative delays between the audio signals, can apply the determined delays to at least one of the audio signals to

form multiple synchronized audio signals, and can mix at least two of the synchronized audio signals in time-varying proportions to form an output audio signal.

[0031] The system **300** can include a housing **302**. In some examples, the housing **302** can be formed from plastic and/or metal. In some examples, the housing **302** can be shaped to fit within a listener's ear.

[0032] The system **300** can include at least one processor **304**. For simplicity, the at least one processor **304** is subsequently referred to as a (single) processor **304**. It will be understood that the (single) processor **304** can alternatively include multiple processors **304** that are in communication with one another. For example, the processor **304** can include a single processor **304** disposed in the housing **302**, multiple processors **304** disposed in the housing **302**, and/or one or more processors **304** disposed in the housing **302** and wirelessly communicating with one or more processors **304** disposed away from the housing **302**.

[0033] A microphone **310** coupled to the processor **304** can convert sound waves proximate the housing **302** to a microphone audio signal that represents an audio content.

[0034] A telecoil **311** coupled to the processor **304** can convert a modulated electromagnetic field proximate the housing **302** to a telecoil audio signal that represents the audio content.

[0035] An antenna **312** coupled to the processor **304** can convert a wireless signal proximate the housing **302** to a wireless audio signal that represents the audio content.

[0036] Although not shown in FIG. 3, other suitable audio signal sources can be used, and can be implemented and connected in a manner similar to the configuration shown in FIG. 3. For example, a vibration sensor coupled to the processor can convert spoken sound waves proximate the vibration sensor to a vibration sensor audio signal that represents a spoken portion of the audio content. For example, the vibration sensor can pick up a listener's contribution to a telephone call (e.g., the portion of the telephone call audio that is spoken by the listener). Other suitable sources can also be used.

[0037] The microphone **310**, the telecoil **311**, and the antenna **312** can be sources for audio signals that represent a same audio content. For example, a movie theater can include speakers that produce audio that can be received by the microphone **310**. The movie theater can also include a room loop around a perimeter of the seating area, which can direct audio to the telecoil **311** via Faraday's law of induction. The movie theater can also include a Bluetooth (or other wireless) transmitter, which can transmit a digital audio signal to the antenna **312**. For this movie theater, the microphone **310**, the telecoil **311**, and the antenna **312** can receive respective audio signals that all represent the audio content in the theater. The system **300** can automatically select and/or combine the audio signals to from the microphone **310**, the telecoil **311**, and the antenna **312** to automatically increase or maximize a signal-to-noise ratio for the listener.

[0038] The system **300** can include memory **306** coupled to the processor **304**. The memory **306** can store instructions that, when executed by the processor **304**, cause the processor **304** to execute operations. The operations are discussed below, and include details regarding how the processor **304** can receive multiple audio signals representing a same audio content, can cross-correlate the multiple audio signals to determine relative delays between the audio signals, can apply the determined delays to at least one of the audio signals to form multiple synchronized audio signals, and can mix at least two of the synchronized audio signals in time-varying proportions to form an output audio signal. A speaker **308** disposed in the housing **302** can produce audio corresponding to the output audio signal. For configurations in which the system **300** is a hearing aid, the housing **302** can position the speaker **308** in a suitable location in a listener's ear.

[0039] In some examples, the operations can include: receiving a first audio signal and a second audio signal that both represent a same audio content; cross-correlating the first and second audio signals to determine a relative delay between the first and second audio signals; applying the determined delay to at least one of the first or second audio signals to form a first synchronized audio signal representing the first audio signal and form a second synchronized audio signal

representing the second audio signal, the first synchronized audio signal being synchronized with the second synchronized audio signal; and mixing the first and second synchronized audio signals in time-varying proportions to form an output audio signal that represents the audio content. The system **300** can include circuitry to perform these operations. The circuitry can include hardware, software (such as executed by the processor **304**), or a combination of hardware and software. FIG. **3** includes a dashed outline to indicate which elements of FIG. **3** can optionally be included in software and executed by the processor **304**. Any or all of these elements can optionally be included in software.

[0040] The microphone **310** can convert sound waves proximate the housing **302** to an analog microphone audio signal. A microphone preamplifier **320** can boost the microphone audio signal to form an analog boosted microphone audio signal. A microphone analog-to-digital converter **330** can convert the boosted microphone audio signal to a digital microphone audio signal. An optional microphone weighted overlap add (WOLA) module **340** can separate the digital microphone audio signal into two or more specified frequency bands, such that downstream processing can be performed for each frequency band, independent of the other frequency bands. At the end of the processing, the frequency band-separated signals can be combined to form a full-frequency audio signal. For simplicity, we continue to refer to a digital microphone audio signal, with the understanding that subsequent processing can be performed on individual frequency bands of the digital microphone audio signal.

[0041] The telecoil **311** can convert a modulated electromagnetic field proximate the housing **302** to an analog telecoil audio signal. A telecoil preamplifier **321** can boost the telecoil audio signal to form an analog boosted telecoil audio signal. A telecoil analog-to-digital converter **331** can convert the boosted telecoil audio signal to a digital telecoil audio signal. An optional telecoil weighted overlap add (WOLA) module **341** can separate the digital telecoil audio signal into two or more specified frequency bands, such that downstream processing can be performed for each frequency band, independent of the other frequency bands. At the end of the processing, the frequency band-separated signals can be combined to form a full-frequency audio signal. For simplicity, we continue to refer to a digital telecoil audio signal, with the understanding that subsequent processing can be performed on individual frequency bands of the digital telecoil audio signal.

[0042] The antenna **312** can convert a wireless signal proximate the housing **302** to an antenna signal. An antenna low-noise amplifier **322** can boost the antenna signal to form a boosted antenna signal. A suitable radio receiver **332** can convert the boosted antenna signal to a digital wireless audio signal. An optional antenna weighted overlap add (WOLA) module **342** can separate the digital wireless audio signal into two or more specified frequency bands, such that downstream processing can be performed for each frequency band, independent of the other frequency bands. At the end of the processing, the frequency band-separated signals can be combined to form a full-frequency audio signal. For simplicity, we continue to refer to a digital wireless audio signal, with the understanding that subsequent processing can be performed on individual frequency bands of the digital wireless audio signal.

[0043] At this stage, the system **300** has produced three digital audio signals. It will be understood that in alternate configurations, two, four, five, or more than five digital audio signals can also be used. The system **300** can use cross-correlation to determine if the three digital audio signals represent the same audio content (by comparing a peak value of a cross-correlation against a specified threshold value). If the system **300** determines that the three audio signals do represent the same audio content, then the cross-correlation can determine relative delay values among the three digital audio signals. The system **300** can use the relative delay values downstream to synchronize the three digital audio signals.

[0044] The system **300** can include a cross-correlation module **350**, **351**, **352** for each of the three digital audio signals. Each cross-correlation module **350**, **351**, **352** can correlate a digital audio signal against one of the other digital audio signals. The output of each cross-correlation module

350, 351, 352 is a set of numerical values that represent a magnitude (or amplitude), as a function of time delay between two of the digital audio signals. Because the two digital audio signals can correspond to the same audio content, the numerical values can show a relatively sharp peak, with the location of the peak representing a relative time delay between the digital audio signals and the peak value representing a strength of the correlation (e.g. a confidence value for the time delay). In some examples, the cross-correlation module **350, 351, 352** can include a comparison with a specified threshold value, such that if the peak value exceeds the threshold, then the system **300** can proceed to use the time delay value downstream. It will be assumed that the peak values exceed the specified threshold, so that the system **300** can continue processing downstream.

[0045] Because updating the relative delays can be computationally intensive, the system **300** may optionally not update the delays in real time. Instead, in some examples, the cross-correlation modules **350, 351, 352** can repeatedly update the relative delays among the audio signals at regularly-spaced intervals, such as every ten seconds, every thirty seconds, every minute, or another suitable time interval, or at irregular intervals. Updating the relative delays relatively infrequently (as opposed to in real time) can reduce the required computation of the processor **304**, which can extend battery life for a hearing aid.

[0046] The system **300** can include multiple cross-correlation modules **350, 351, 352**, such as one cross-correlation module for each digital audio signal (e.g., microphone, telecoil, and antenna). Each digital audio signal can be fed to two of the cross-correlation modules **350, 351, 352**, so that the cross-correlation modules **350, 351, 352**, together, can calculate the relative delay of any digital audio signal relative to any other digital audio signal.

[0047] In some examples, for a particular audio signal, the system **300** can determine that the absolute peak value of the cross-correlation is negative. Based on the absolute peak value being negative, the system **300** can determine that the two corresponding audio signals are out of phase with each other. To correct for this phase difference, the system **300** can invert one of the two audio signals to bring the two audio signals into phase with each other. The system **300** can apply this phase correction to all of the audio signals, such that the synchronized audio signals become in phase (specifically, not 180 degrees out of phase) with one another.

[0048] A microphone delay **360** can receive relative delay values from two or more of the cross-correlation modules **350, 351, 352**, and can delay the digital microphone audio signal by a suitable delay (including zero if the digital microphone audio signal leads the other two digital audio signals) to form a synchronized microphone audio signal.

[0049] A telecoil delay **361** can receive relative delay values from two or more of the cross-correlation modules **350, 351, 352**, and can delay the digital telecoil audio signal by a suitable delay (including zero if the digital telecoil audio signal leads the other two digital audio signals) to form a synchronized telecoil audio signal.

[0050] An antenna delay **362** can receive relative delay values from two or more of the cross-correlation modules **350, 351, 352**, and can delay the digital wireless audio signal by a suitable delay (including zero if the digital wireless audio signal leads the other two digital audio signals) to form a synchronized wireless audio signal.

[0051] At this stage, the system **300** has now produced multiple synchronized audio signals that are synchronized with one another. The system **300** can now combine one or more of the synchronized audio signals to form an output audio signal, and to increase or maximize a signal-to-noise ratio of the output audio signal.

[0052] At amplifiers (or attenuators) **370, 371, and 372**, and combiner **380**, the system **300** can mix the synchronized audio signals in time-varying proportions, in real time, to increase or maximize a signal-to-noise ratio of the output audio signal **390**. Mixing in time-varying proportions, in real time, can include, repeatedly: adjusting the proportions of the synchronized audio signals in the output audio signal and determining a signal-to-noise ratio of the output audio signal, until the signal-to-noise ratio has approached a maximum value with respect to the proportions of the

synchronized audio signals in the output audio signal. In some examples, the synchronized audio signals each have a corresponding signal-to-noise ratio, and the output audio signal **390** has a signal-to-noise ratio that is greater than or equal to the signal-to-noise ratios of the synchronized audio signals. In some examples, the system **300** can use dithering to adjust the proportions of the synchronized audio signals in the output audio signal. In some examples, the system **300** can use a hill-climbing algorithm to adjust the proportions of the synchronized audio signals in the output audio signal.

[0053] There are numerous ways to determine a signal-to-noise ratio of an audio signal, in real time or near-real time. For example, for speech, a signal-to-noise ratio can scale inversely with a sound level between syllables in the speech, or the sound level between syllables scaled by a sound level during the syllables. Additional examples are provided in the article “Estimation of Signal-to-Noise Ratios in Realistic Sound Scenarios,” by Karolina Smeds, Florian Wolters, and Martin Rung, Journal of the American Academy of Audiology, Vol. 26, No. 2, 2015, pp. 183-196, which is incorporated by reference herein in its entirety. Other suitable techniques can also be used to determine the signal-to-noise ratio of the output audio signal.

[0054] FIG. **4** shows a flow chart of a method **400** for generating an audio signal from multiple inputs, in accordance with some examples. The method **400** can be executed on the system **100** of FIG. **1**, the system **300** of FIG. **3**, or on any suitable system. The method **400** is but one example of a method for generating an audio signal from multiple inputs. Other suitable methods can also be used.

[0055] At operation **402**, the system can receive a first audio signal and a second audio signal that both represent a same audio content.

[0056] At operation **404**, the system can cross-correlate the first and second audio signals to determine a relative delay between the first and second audio signals.

[0057] At operation **406**, the system can apply the determined delay to at least one of the first or second audio signals to form a first synchronized audio signal representing the first audio signal and form a second synchronized audio signal representing the second audio signal, the first synchronized audio signal being synchronized with the second synchronized audio signal.

[0058] At operation **408**, the system can mix the first and second synchronized audio signals in time-varying proportions to form an output audio signal that represents the audio content.

[0059] In some examples, operation **408** can optionally further include mixing the first and second synchronized audio signals in time-varying proportions, in real time, to increase or maximize a signal-to-noise ratio of the output audio signal.

[0060] In some examples, operation **408** can optionally further include, repeatedly: adjusting the proportions of the first and second synchronized audio signals in the output audio signal; and determining a signal-to-noise ratio of the output audio signal, until the signal-to-noise ratio has approached a maximum value with respect to the proportions of the first and second synchronized audio signals in the output audio signal.

[0061] In some examples, operation **404** can optionally further include: determining that a correlation of the first and second audio signals has an absolute peak value that exceeds a specified correlation value threshold; and determining the relative delay from a location of the absolute peak value.

[0062] In some examples, operation **404** can optionally further include: determining that the absolute peak value is negative; determining, based on the absolute peak value being negative, that the first audio signal is out of phase with the second audio signal; and inverting one of the first audio signal or the second audio signal, such that the first and second synchronized audio signals are in phase.

[0063] In some examples, method **400** can optionally further include: spectrally decomposing the first audio signal into a specified plurality of adjoining frequency bands to form a plurality of first audio channels; spectrally decomposing the second audio into the specified plurality of adjoining

frequency bands to form a plurality of second audio channels; for each frequency band: cross-correlating the first and second audio channels to determine a relative delay between the first and second audio channels; applying the determined delay to at least one of the first or second audio channels to form a first synchronized audio channel representing the first audio channel and form a second synchronized audio channel representing the second audio channel, the first synchronized audio channel being synchronized with the second synchronized audio channel; and mixing the first and second synchronized audio channels in time-varying proportions to form an output audio channel that represents the audio content in the frequency band; and combining the output audio channels to form the output audio signal.

[0064] There are additional options that can be used with the systems **100**, **300** and the method **400**.

[0065] In some examples, a leading audio signal can be delayed by a number of samples corresponding to a peak in the cross-correlation curve, but the rate of cross-correlation operations can increase during this adaptation to gain confidence that this delay value is correct or optimal. The chosen time shifts from the past several cross-correlation operations can be averaged together (e.g., through exponential averaging) to determine the best time shift or delay to apply to the leading signal. In general, the time shift can be dynamic to account for unknown processing delay in the loop system.

[0066] In some examples, the microphone signal can be designated as a primary signal, such that the phase of the other audio signals can be inverted if needed to match a phase of the microphone signal.

[0067] In some examples, if a Boolean trigger event occurs (e.g. a magnetic switch activates or a wireless stream is detected), a hearing aid can respond by switching to the corresponding input (e.g. telecoil for autocoil, microphone for autophone, wireless audio for stream, and so forth). While in this mode, the hearing aid can perform periodic cross-correlation operations with other appropriate inputs and can be used to increase or maximize a signal-to-noise ratio of the output audio signal.

[0068] In some examples, the system can change which input audio signals are monitored, depending on which input audio signals is the primary input. For example, if the autocoil is the primary input, the system can monitor the microphone signal. If the autophone is the primary input, the system can monitor the telecoil signal. If the stream is the primary input, the system can monitor the microphone signal and the telecoil signal. Other combinations are also possible. In some examples, the appropriate inputs can vary situationally. For example, if the hearing aid can tell the difference between a public/broadcast stream (e.g., an airport announcement) and a private/addressed stream (e.g., a phone call), the system can vary which input audio signals are monitored as needed.

[0069] In some examples, the system can force the lowest channel in directional mode to stay omnidirectional. This can reduce hearing aid self-noise for a greater signal-to-noise improvement than attempting to stay directional.

[0070] In some examples, the system can apply heuristics to make some caveats to its algorithm. For example, when the acoustic signal is correlated to a low-fidelity stream, the system can weight the highest frequencies 100% to the microphone signal, even though it may not correlate to the stream at those frequencies because the low-fidelity stream may not extend to frequencies that high. There can be other circumstances in which the system selects a wider bandwidth over an improved narrowband signal-to-noise ratio.

[0071] In some examples, the system can apply heuristics regarding a maximum value of delay. If there is a streamed signal present, the system can automatically switch to the streamed input. If the cross-correlation determines that the acoustic signal is highly correlated to the stream, and has a much shorter delay, the system can begin to mix the acoustic signal and the stream to increase or maximize the signal-to-noise ratio of the output. In some examples, if the mixing weights are close to 50%, or even skew toward favoring the microphone signal, the system can decide to no longer delay the microphone signal for mixing, can decide to ignore the stream, and can use only the

microphone signal to reduce the total audio delay.

[0072] In some examples, the system can apply additional heuristics when band-by-band signal-to-noise ratio optimization may not make sense. For instance, if we know that a particular input signal has a limited bandwidth (e.g., due to streaming codec limitations), the high-frequency signal may not be correlated. For these situations, if low-frequency signals turn out to be correlated, the system can weight the secondary correlated signal 100% in the high bands to restore high frequency content.

[0073] In some examples, the system can monitor a signal level (e.g., a root-mean-square signal level) in band to detect when an input has a limited bandwidth. If several of the highest-frequency bands have relatively low levels, such as below a specified threshold for a specified duration, the system can conclude that the signal source does not have any content at those higher frequencies. This can free the system from cataloguing the bandwidths of the various types of signal and can allow the system to work with unknown sources that have unknown bandwidths.

[0074] In some examples, the system can limit use of an audio signal at particular frequencies or frequency bands. For example, for a telephone call, a listener would not expect to hear any content at frequencies higher than about 3 kHz or 4 kHz. As a result, the system can block content for frequencies above about 3 kHz or 4 kHz, for a phone input, including a stream or a telecoil. This can help avoid having different-sounding content in adjacent frequency bands. Similarly, a vibration sensor may only have content below a threshold frequency, such as about 1 kHz. The system can optionally block content about 1 kHz for the vibration sensor signal.

[0075] In some examples, the system can allow a listener to access settings of this signal processing, such as through a mobile application that runs on a user device, such as a smart phone. In this manner, the listener can manually adjust a mix, and/or adjust preferences for how audio signals are mixed, including parameters such as a speed and priority. In some examples, the mobile application can notify the listener that a better-sounding input is available. In some examples, an audible alert can notify the listener that a better-sounding input is available.

[0076] In some examples, the system can include leveling logic to maintain suitable signal levels in each band.

[0077] In some examples, the system can additionally include features that allow audio to be presented to a listener in both ears, rather than a single ear. Such features can help ensure that audio content can be balanced between a listener's ears.

[0078] Although the inventive concept has been described in detail for the purpose of illustration based on various examples, it is to be understood that such detail is solely for that purpose and that the inventive concept is not limited to the disclosed examples, but, on the contrary, is intended to cover modifications and equivalent arrangements that are within the spirit and scope of the appended claims. For example, it is to be understood that the present disclosure contemplates that, to the extent possible, one or more features of any example can be combined with one or more features of any other example.

[0079] Furthermore, since numerous modifications and changes will readily occur to those with skill in the art, it is not desired to limit the inventive concept to the exact construction and operation described herein. Accordingly, all suitable modifications and equivalents should be considered as falling within the spirit and scope of the present disclosure.

Examples

[0080] To further illustrate the device, related system, and/or related method discussed herein, a non-limiting list of examples is provided below. Each of the following non-limiting examples can stand on its own, or can be combined in any permutation or combination with any one or more of the other examples.

[0081] In Example 1, a system for generating an audio signal from multiple inputs system can include: at least one processor; and memory coupled to the at least one processor, the memory configured to store instructions that, when executed by the at least one processor, cause the at least

one processor to execute operations, the operations comprising: receiving a first audio signal and a second audio signal that both represent a same audio content; cross-correlating the first and second audio signals to determine a relative delay between the first and second audio signals; applying the determined delay to at least one of the first or second audio signals to form a first synchronized audio signal representing the first audio signal and form a second synchronized audio signal representing the second audio signal, the first synchronized audio signal being synchronized with the second synchronized audio signal; and mixing the first and second synchronized audio signals in time-varying proportions to form an output audio signal that represents the audio content.

[0082] In Example 2, the system of Example 1 can optionally be configured such that the operations further comprise: mixing the first and second synchronized audio signals in time-varying proportions, in real time, to increase or maximize a signal-to-noise ratio of the output audio signal.

[0083] In Example 3, the system of any one of Examples 1-2 can optionally be configured such that mixing the first and second synchronized audio signals in time-varying proportions, in real time, to increase or maximize the signal-to-noise ratio of the output audio signal comprises, repeatedly: adjusting the proportions of the first and second synchronized audio signals in the output audio signal; and determining a signal-to-noise ratio of the output audio signal, until the signal-to-noise ratio has approached a maximum value with respect to the proportions of the first and second synchronized audio signals in the output audio signal.

[0084] In Example 4, the system of any one of Examples 1-3 can optionally be configured such that the first and second audio signals each have a corresponding signal-to-noise ratio; and the output audio signal has a signal-to-noise ratio that is greater than or equal to the signal-to-noise ratios of the first and second audio signals.

[0085] In Example 5, the system of any one of Examples 1-4 can optionally be configured such that the operations further comprise: cross-correlating the first and second audio signals, repeatedly, to update the relative delay between the first and second audio signals.

[0086] In Example 6, the system of any one of Examples 1-5 can optionally be configured such that cross-correlating the first and second audio signals to determine the relative delay between the first and second audio signals comprises: determining that a correlation of the first and second audio signals has an absolute peak value that exceeds a specified correlation value threshold; and determining the relative delay from a location of the absolute peak value.

[0087] In Example 7, the system of any one of Examples 1-6 can optionally be configured such that the operations further comprise: determining that the absolute peak value is negative; determining, based on the absolute peak value being negative, that the first audio signal is out of phase with the second audio signal; and inverting one of the first audio signal or the second audio signal, such that the first and second synchronized audio signals are in phase.

[0088] In Example 8, the system of any one of Examples 1-7 can optionally be configured such that the operations further comprise: spectrally decomposing the first audio signal into a specified plurality of adjoining frequency bands to form a plurality of first audio channels; spectrally decomposing the second audio into the specified plurality of adjoining frequency bands to form a plurality of second audio channels; for each frequency band: cross-correlating the first and second audio channels to determine a relative delay between the first and second audio channels; applying the determined delay to at least one of the first or second audio channels to form a first synchronized audio channel representing the first audio channel and form a second synchronized audio channel representing the second audio channel, the first synchronized audio channel being synchronized with the second synchronized audio channel; and mixing the first and second synchronized audio channels in time-varying proportions to form an output audio channel that represents the audio content in the frequency band; and combining the output audio channels to form the output audio signal.

[0089] In Example 9, the system of any one of Examples 1-8 can optionally be configured such that

the operations further comprise, for each frequency band: adjusting a volume level of each output audio channel to a specified volume level.

[0090] In Example 10, the system of any one of Examples 1-9 can optionally further include a microphone coupled to the at least one processor and configured to convert sound waves proximate the microphone to a microphone audio signal, the microphone audio signal being the first audio signal.

[0091] In Example 11, the system of any one of Examples 1-10 can optionally further include a telecoil coupled to the at least one processor and configured to convert a modulated electromagnetic field proximate the telecoil to a telecoil audio signal, the telecoil audio signal being the second audio signal.

[0092] In Example 12, the system of any one of Examples 1-11 can optionally further include an antenna coupled to the at least one processor and configured to convert a wireless signal proximate the antenna to an wireless audio signal, the wireless audio signal being the second audio signal.

[0093] In Example 13, a method for generating an audio signal from multiple inputs can include: receiving a first audio signal and a second audio signal that both represent a same audio content; cross-correlating the first and second audio signals to determine a relative delay between the first and second audio signals; applying the determined delay to at least one of the first or second audio signals to form a first synchronized audio signal representing the first audio signal and form a second synchronized audio signal representing the second audio signal, the first synchronized audio signal being synchronized with the second synchronized audio signal; and mixing the first and second synchronized audio signals in time-varying proportions to form an output audio signal that represents the audio content.

[0094] In Example 14, the method of Example 13 can optionally further include: mixing the first and second synchronized audio signals in time-varying proportions, in real time, to increase or maximize a signal-to-noise ratio of the output audio signal.

[0095] In Example 15, the method of any one of Examples 13-14 can optionally be configured such that mixing the first and second synchronized audio signals in time-varying proportions, in real time, to increase or maximize the signal-to-noise ratio of the output audio signal comprises, repeatedly: adjusting the proportions of the first and second synchronized audio signals in the output audio signal; and determining a signal-to-noise ratio of the output audio signal, until the signal-to-noise ratio has approached a maximum value with respect to the proportions of the first and second synchronized audio signals in the output audio signal.

[0096] In Example 16, the method of any one of Examples 13-15 can optionally further include: cross-correlating the first and second audio signals, repeatedly, to update the relative delay between the first and second audio signals.

[0097] In Example 17, the method of any one of Examples 13-16 can optionally be configured such that cross-correlating the first and second audio signals to determine the relative delay between the first and second audio signals comprises: determining that a correlation of the first and second audio signals has an absolute peak value that exceeds a specified correlation value threshold; and determining the relative delay from a location of the absolute peak value.

[0098] In Example 18, the method of any one of Examples 13-17 can optionally further include: determining that the absolute peak value is negative; determining, based on the absolute peak value being negative, that the first audio signal is out of phase with the second audio signal; and inverting one of the first audio signal or the second audio signal, such that the first and second synchronized audio signals are in phase.

[0099] In Example 19, the method of any one of Examples 13-18 can optionally further include: spectrally decomposing the first audio signal into a specified plurality of adjoining frequency bands to form a plurality of first audio channels; spectrally decomposing the second audio into the specified plurality of adjoining frequency bands to form a plurality of second audio channels; for each frequency band: cross-correlating the first and second audio channels to determine a relative

delay between the first and second audio channels; applying the determined delay to at least one of the first or second audio channels to form a first synchronized audio channel representing the first audio channel and form a second synchronized audio channel representing the second audio channel, the first synchronized audio channel being synchronized with the second synchronized audio channel; and mixing the first and second synchronized audio channels in time-varying proportions to form an output audio channel that represents the audio content in the frequency band; and combining the output audio channels to form the output audio signal.

[0100] In Example 20, an ear-wearable device can include: a housing; at least one processor disposed in the housing; a microphone coupled to the at least one processor and configured to convert sound waves proximate the housing to a microphone audio signal that represents an audio content; a telecoil coupled to the at least one processor and configured to convert a modulated electromagnetic field proximate the housing to a telecoil audio signal that represents the audio content; an antenna coupled to the at least one processor and configured to convert a wireless signal proximate the housing to a wireless audio signal that represents the audio content; and memory coupled to the at least one processor, the memory configured to store instructions that, when executed by the at least one processor, cause the at least one processor to execute operations, the operations comprising: receiving the microphone audio signal, the telecoil audio signal, and the wireless audio signal; cross-correlating the microphone audio signal, the telecoil audio signal, and the wireless audio signal to determine relative delays among the microphone audio signal, the telecoil audio signal, and the wireless audio signal; applying the determined delays to at least some of the microphone audio signal, the telecoil audio signal, or the wireless audio signal to form a synchronized microphone audio signal representing the microphone audio signal, form a synchronized telecoil audio signal representing the telecoil audio signal, and form a synchronized wireless audio signal representing the wireless audio signal, the synchronized microphone audio signal, the synchronized telecoil audio signal, and the synchronized wireless audio signal being synchronized to one another; and mixing at least two of the synchronized microphone audio signal, the synchronized telecoil audio signal, or the synchronized wireless audio signal in time-varying proportions to form an output audio signal that represents the audio content; and a speaker disposed in the housing and configured to produce audio corresponding to the output audio signal.

Claims

1. A system for generating an audio signal from multiple inputs, comprising: at least one processor; and memory coupled to the at least one processor, the memory configured to store instructions that, when executed by the at least one processor, cause the at least one processor to execute operations, the operations comprising: spectrally decomposing a plurality of input signals into a plurality of frequency bands; monitoring signal levels across the plurality of frequency bands to detect bandwidth limitations in at least one of the plurality of input signals; determining mixing weights for a subset of the plurality of frequency bands based on the bandwidth limitations; and generating a bandwidth-enhanced output signal by applying the mixing weights to combine corresponding frequency bands of the plurality of input signals.
2. The system of claim 1, wherein monitoring signal levels further includes monitoring root mean square (RMS) levels in each frequency band over time to automatically detect when an input signal has limited bandwidth.
3. The system of claim 1, wherein determining mixing weights further includes weighting a second input signal at substantially 100% in high frequency bands where bandwidth limitations are detected in a first input signal.
4. The system of claim 1, wherein generating the bandwidth-enhanced output signal further includes: identifying a set of bandwidth-deficient frequency bands based on the bandwidth limitations; and selectively applying the mixing weights to restore signal content in the set of

bandwidth-deficient frequency bands.

5. The system of claim 1, wherein monitoring signal levels further includes: identifying a low signal level state, the low signal level state indicating that a subset of frequency bands have consistently low signal levels over a predetermined duration; and determining that an input signal has limited bandwidth in response to identifying the low signal level state.
6. The system of claim 5, wherein the operations further include: determining a signal prioritization, the signal prioritization identifying whether to prioritize wider bandwidth or improved narrowband signal quality for specific frequency bands based on identifying the low signal level state; and adjusting the mixing weights based on the signal prioritization.
7. The system of claim 1, wherein the operations further include: detecting an input signal type; selecting enhancement parameters based on the input signal type; and applying the enhancement parameters while maintaining signal quality across frequency bands.
8. The system of claim 1, wherein the operations further include: maintaining signal levels on a per-channel basis when combining the plurality of frequency bands; and maintaining consistent signal levels across frequency band boundaries to substantially prevent switching artifacts.
9. The system of claim 1, wherein determining mixing weights further includes: analyzing signal quality in each frequency band independently; adjusting the mixing weights on a channel-by-channel basis; and optimizing the mixing weights to maximize signal quality.
10. The system of claim 1, wherein: a first input signal further includes a limited bandwidth signal from a companion microphone; and generating the bandwidth-enhanced output signal further includes weighting a hearing aid microphone signal more heavily at higher frequencies even when the plurality of input signals are not correlated at those frequencies.
11. The system of claim 1, wherein: a first input signal is received from a vibration sensor; and generating the bandwidth-enhanced output signal further includes frequency-specific mixing for at least one of voice enhancement or occlusion reduction.
12. The system of claim 1, wherein: a first input signal is received from a loop system; and generating the bandwidth-enhanced output signal further includes applying more weight to an acoustic microphone signal at lower frequencies.
13. The system of claim 1, wherein the operations further include forcing specified frequency bands to work together for specific input types by applying consistent mixing weights across the specified frequency bands.
14. The system of claim 13, wherein forcing specified frequency bands to work together further includes limiting phone audio processing to frequencies below approximately 4 kHz.
15. A method for generating an audio signal from multiple inputs, comprising: spectrally decomposing a plurality of input signals into a plurality of frequency bands; monitoring signal levels across the plurality of frequency bands to detect bandwidth limitations in at least one of the plurality of input signals; determining mixing weights for a subset of the plurality of frequency bands based on the bandwidth limitations; and generating a bandwidth-enhanced output signal by applying the mixing weights to combine corresponding frequency bands of the plurality of input signals.
16. The method of claim 15, wherein determining mixing weights comprises weighting a second input signal at substantially 100% in high frequency bands where bandwidth limitations are detected in a first input signal.
17. The method of claim 15, wherein generating the bandwidth-enhanced output signal comprises: identifying a set of bandwidth-deficient frequency bands based on the bandwidth limitations; and selectively applying the mixing weights to restore signal content in the set of bandwidth-deficient frequency bands.
18. The method of claim 15, wherein monitoring signal levels comprises: identifying a low signal level state, the low signal level state indicating that a subset of frequency bands have consistently low signal levels over a predetermined duration; and determining that an input signal has limited

bandwidth in response to identifying the low signal level state.

19. The method of claim 18, further comprising: determining a signal prioritization, the signal prioritization identifying whether to prioritize wider bandwidth or improved narrowband signal quality for specific frequency bands based on identifying the low signal level state; and adjusting the mixing weights based on the signal prioritization.

20. The method of claim 15, further comprising: detecting an input signal type; selecting enhancement parameters based on the input signal type; and applying the enhancement parameters while maintaining signal quality across frequency bands.
